

QLOT: Queried Learned Optimization for Face Landmark Tracking

Supplementary Material

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1. Ablations

We evaluate QLOT against a set of pre-specified claims, each paired with a controlled counterfactual that changes exactly one component relative to the full model (*QLOT-full*). All ablations are trained from scratch with the same data mixture, augmentations, optimizer, learning-rate schedule, iteration curriculum, and checkpoint-selection rule. Except for A2’s approximately parameter-matched local updater, where a component is removed, the freed capacity is left un-replaced rather than padded with additional parameters. Parameter counts, rounded to the nearest 0.01M, are reported alongside. The claims and their tests are:

- C1 Efficiency-accuracy-stability.** QLOT attains competitive landmark accuracy with substantially lower model cost and improved temporal stability relative to direct coordinate-regression baselines. **Test:** external comparison under a matched protocol (accuracy, temporal stability, parameters, MACs).
- C2 Recurrent memory.** Carrying a recurrent hidden state across frames improves temporal stability beyond coordinate initialization alone. **Test:** (A1, no hidden state carry) reset the hidden state to zero at every new frame while still prefilling the previous frame’s predicted coordinates and covariance, during training and evaluation.
- C3 Update operator.** The GRU-like gated update with lightweight Write-Mix-Read (WMR) communication performs at least as well as PyTorch’s *GRUCell* updater, a *local-only* updater, and a *mean-only* WMR variant. **Tests:** (A2) approximately parameter-matched per-landmark GRU (our variant) while removing the WMR block; (A3) a standard GRU in place of our GRU variant; (A4) WMR with the dispersion descriptor and slot-energy decomposition removed.
- C4 Uncertainty supervision.** A full 2D covariance output with spatial and temporal GNLL losses reduces jitter, without harming accuracy, over a simple L_2 coordinate loss. **Tests:** (A5) covariance head and all GNLL terms

removed, trained with L_2 only; (A6) temporal GNLL terms removed, spatial GNLL retained.

- C5 Query encoding.** The modified Phase-Modulated Positional Encoding (PMPE) improves performance over a standard geometric-progression Fourier encoding. **Test:** (A7) swap the PMPE-variant for the Fourier-only encoding of [1] at matched width.

A1 isolates recurrent **memory** (not tracker initialization, since coordinates are still carried); A2–A4 isolate the **update operator** under identical feature extractors, hidden-state width, and training budget; A5–A6 isolate the **supervision recipe**; A7 isolates the **query encoding**. This yields one reference plus seven single-change ablations (eight configurations), each targeting one claim.

Conventions. All ablations report means over 3 seeds on identical data manifests. Significance is assessed with a paired bootstrap (10,000 resamples) over pooled seed-video pairs for temporal metrics and pooled seed-image pairs for static NME; n.s. (not significant) indicates $p > 0.05$. Static NME is inter-ocular (distance between outer eye corners) normalized on WFLW/300-W and face-size (square-root of label bounding box area) normalized on FaceSynthetics. WFLW-V metrics are face-size normalized on the 150-clip test split (75 easy / 75 hard), 4 refinement iterations at frame 0, and 1 thereafter.

Checkpoint selection rule. Among the top-3 validation candidates ordered by NME, select the one with the lowest validation NMF.

1.1. Results

Quantitative results are in Table 1 and Table 2.

1.1.1. Temporal memory enables one-iteration-per-frame tracking (C2)

A central design claim is that QLOT tracks with a single refinement iteration per frame because it carries a recurrent hidden state, not merely the previous coordinates. Ablation A1 tests this directly. It is architecturally identical to the full model but resets the hidden state to zero at every new video

Config	Params	WFLW	300-W	Synth	V-NME	NMF _E	NMF _H	NAD(1)	NAD(4)	NAD(16)
A0 (QLOT-full)	2.01M	3.934	3.487	1.617	1.089	78.97	140.02	3.02	4.14	5.15
A1 (no state carry)	2.01M	3.938	3.498	1.646 [†]	1.825 [†]	170.72 [†]	252.11 [†]	5.84 [†]	10.37 [†]	11.83 [†]
A2 (local-only)	1.95M	4.277 [†]	3.626 [†]	1.767 [†]	1.425 [†]	85.14 [†]	156.83 [†]	3.34 [†]	4.77 [†]	6.35 [†]
A3 (standard GRU)	2.01M	3.947	3.431 [†]	1.646 [†]	1.107 [†]	78.67	139.62	3.01	4.13	5.19
A4 (mean-only WMR)	1.98M	3.934	3.493	1.619	1.066 [†]	77.69 [†]	138.52 [†]	2.99 [†]	4.07 [†]	5.08 [†]
A5 (L_2 only)	1.99M	3.918	3.493	1.611 [†]	1.088	80.47 [†]	144.16 [†]	3.10 [†]	4.19 [†]	5.18
A6 (no temporal GNLL)	2.01M	3.914	3.505 [†]	1.597 [†]	1.095	80.34 [†]	143.95 [†]	3.10 [†]	4.20 [†]	5.17
A7 (Fourier PE)	2.03M	3.930	3.501	1.617	1.092	78.00 [†]	140.52	3.02	4.13	5.13

Table 1. Ablation results, mean over 3 seeds. WFLW/300-W: static NME (% , inter-ocular); Synth: FaceSynthetics static NME (% , face-size); V-NME: WFLW-V video NME (% , face-size); NMF_E / NMF_H: NMF on the 75 easy / 75 hard WFLW-V test clips truncated to 120 frames; NAD(m) values are scaled by 10^3 . Bold marks the best point estimate in each metric column. [†] marks a significant difference vs. A0 from paired bootstrap.

frame while still receiving the previous frame’s predicted coordinates and covariance as the starting estimate.

The effect is large and unambiguous. On WFLW-V, A1’s normalized mean flicker nearly doubles (211.4 vs. 109.5, +93%, $p < 0.05$ paired), video NME rises by 68% (1.83% vs. 1.09%), and NAD is elevated across the entire measured range (NAD(1) +93%, NAD(4) +150%, NAD(16) +130%). Static NME is unchanged on WFLW and 300-W (3.94% vs. 3.93% and 3.50% vs. 3.49%, respectively; both n.s.), while FaceSynthetics shows a small but significant +1.8% degradation. At frame 0, where both models run four refinement iterations, the two are indistinguishable (1.41% vs. 1.42% NME, n.s.). The gap opens at frame 1, the first one-iteration frame, and persists until the end of the 120-frame clips (results are averaged over clips). The hidden state is therefore not a smoother applied on top of an otherwise-converged tracker. It is what allows the tracker to stay locked in one frame, or when lost, recover more quickly. Coordinate prefill tells the model *where* to focus in the next frame, the hidden state carries *what the track has learned* about appearance, motion, and local evidence.

1.1.2. Cross-landmark communication and the update operator (C3)

We ablate the update operator along three axes. Removing cross-landmark communication while expanding the local GRU to approximately match A0’s parameter count (A2) is the most damaging architectural change after A1: NMF +10.5%, video NME +31%, and WFLW NME +8.7% (4.28% vs. 3.93%), all significant.

Replacing our GRU-variant with a standard GRU (A3, parameter-matched) leaves jitter unchanged (NMF −0.3%, n.s.) and video NME marginally worse (+1.7%, small but significant in the pooled comparison), while slightly *improving* static 300-W accuracy (−1.6%, significant). Thus, our GRU variant matches the standard recurrent cell on

NMF and NAD, while A3 incurs a small video-NME cost and improves 300-W accuracy; a broad advantage for either update formulation is not demonstrated.

1.1.3. The dispersion descriptor is unnecessary (C3)

The WMR mixer’s slot representation includes a dispersion descriptor in addition to the weighted mean. Removing it as well as the slot-energy decomposition (A4) while keeping routing, slot attention, and the gated update the same does not degrade any metric. To the contrary, temporal stability is *improved*: NMF −1.3% and video NME −2.1% (both significant across seeds), NAD(1)/(4)/(16) all lower, with static accuracy unchanged (WFLW 3.934% vs. 3.934%, n.s.).

We theorize that the spread information in the dispersion is compensated by the routed temporal states. Additionally, noise susceptibility is increased since variance computation is a high-pass filter. Other applications adopting the WMR block, however, may still benefit from the slot disagreement and normalization mechanisms.

We adopt the simpler mean-only WMR, and call this configuration **QLOT-final**. It is 1.7% smaller, marginally cheaper in MACs, and measurably more stable.

1.1.4. Supervision: Temporal GNLL drives stability (C4)

Training with a full 2D covariance and spatial+temporal GNLL losses reduces jitter at no accuracy cost. Removing the temporal GNLL terms (A6) raises NMF by 2.4% and NAD(1)/(4) significantly, while static effects are small and mixed: WFLW is unchanged, FaceSynthetics improves by 1.2% significantly, and a +0.5% shift on 300-W barely reaches significance with inconsistent signs across seeds; removing the covariance head and all GNLL terms in favor of a pure L_2 coordinate loss (A5) produces a nearly identical +2.6% NMF, indicating the temporal terms account for essentially all of the gain. The effect is confined to short timescales (NAD(16) is unaffected for both A5 and A6).

A1 checkpoints	300-W	WFLW	V-NME	NMF _e	NMF _h	NAD(4)	NAD(16)
stability-aware (A1)	3.498	3.938	1.825	170.72	252.11	10.37	11.83
NME-only (A1-nme)	3.485 [†]	3.938	1.861 [†]	172.16 [†]	254.10 [†]	10.52 [†]	12.08 [†]

Table 2. Effect of the checkpoint-selection rule on the A1 ablation, mean over 3 seeds (columns as in Table 1, NAD scaled by 10^3). Both rows are the **same three training runs**; only the selected checkpoint differs. [†] marks a significant difference between the two selection rules from paired bootstrap. Static accuracy is identical on WFLW, while all temporal metrics degrade under NME-only selection, on both the easy and hard subsets.

For context, we place these NMF values next to the strongest published baseline. RWR [2] reports an NMF of 127.9 on WFLW-V (82.7 easy / 173.0 hard), already below every EMA-smoothed detector it was compared against, whereas A5 and A6 attain 112.3 and 112.1 (80.5/80.3 easy, 144.2/144.0 hard) and QLOT-full 109.5 (79.0/140.0)—12–14% lower overall, with the largest margin on the hard split (19% vs. 4–5% on easy).

The test protocols are close enough to support this reading. The metric and the official easy/hard video partition are identical, our 150 held-out test videos are a stratified draw from the same 1,000-video pool that RWR evaluates in full, and our fixed 120-frame windows cover all but at most the final 31 frames of each 120–151-frame video. RWR’s whole-dataset evaluation averages over more data from the same distribution. The remaining difference lies in training, not testing, and is architectural. RWR is a purely per-frame model, so it can only be trained on static images. Our models, by contrast, train on WFLW-V clips from videos disjoint from the test set, so part of the gap may reflect in-domain video adaptation that RWR structurally cannot exploit, rather than the tracking method alone. Subject to that caveat, we read the comparison as QLOT being measurably more stable than the strongest published baseline under near-identical test conditions, rather than as a fully controlled head-to-head.

Within our own protocol-matched ablations, the temporal-GNLL effect is real but an order of magnitude smaller than the state-carry effect (C2), +2.4% NMF for A6 vs. +93% for A1.

1.1.5. Query Encoding (C5)

Replacing the PMPE-variant with a geometric-progression Fourier encoding (A7) yields no measurable difference in aggregate metrics. Easy-split NMF improves slightly (78.00 vs. 78.97, significant), without a corresponding hard-split or overall effect. We therefore find no aggregate benefit for either encoding and retain PMPE for its flexibility.

1.1.6. Checkpoint Selection

We found checkpoint selection to be a first-class methodological variable for tracking. To quantify it, we re-selected the A1 checkpoints purely by validation NME (dropping the minimal-NMF-of-top-3 rule) and evaluated them on the test set (Table 2).

NME-only checkpoints are worse on every temporal metric (NMF 213.1 vs. 211.4, +0.8% overall, +0.8% on easy and +0.8% on hard; video NME +1.9%; NAD higher at all timescales; all significant) while WFLW is unchanged and 300-W improves slightly (3.485 vs. 3.498, significant).

1.2. Model Complexity

QLOT-final has 1.98M parameters. Its computational cost depends on the number of refinement iterations I run per frame: 872.79M MACs per frame for $I = 4$ iterations (used at frame 0 of each clip) and 648.87M MACs per frame for $I = 1$ (used for all subsequent frames), estimated from the architecture. The backbone, FPN, correlation volume generation, and query encoding run only once per frame. The marginal cost of each additional refinement iteration is about 75M MACs. The four-iteration initialization is about 1.35 times the steady-state per-frame cost. All MAC counts are at 224×224 input resolution and 98 landmarks. Figure 1 shows the inference time plotted against the number of tracked landmarks.

2. Reproducibility

The code uses AR(1) dynamics correction for the delta GNLL loss with $\rho = 0.74$. This is a remnant of an earlier implementation in which the coordinates were detached instead of the covariance. Since the covariance is now detached (and the coordinates live), the correction is not necessary. ρ can be absorbed into the loss weight (so the effective weight is larger with AR(1) correction).

We use the HGNetV2 implementation from the `timm` (PyTorch Image Models) library [3] as the backbone.

3. Analysis of the Normalized Allan Deviation

This section derives the NAD signatures used in the *Temporal Stability Metrics* section, gives finite-range slope benchmarks, the estimation protocol, and the fitted results.

3.1. Setup and Exact Identities

Recall $\tilde{e}_{ln}$ is the face-size-normalized landmark error for landmark n in frame l ,

$$\tilde{e}_{ln} = (\mathbf{p}_{ln} - \mathbf{y}_{ln})/d_l, \quad (1)$$

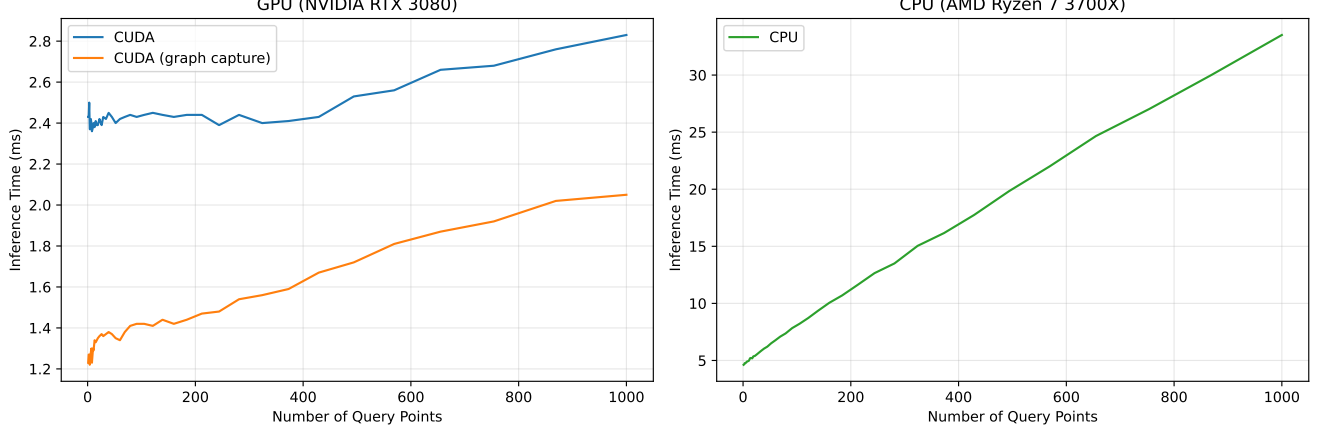


Figure 1. Inference time of QLOT via ONNX runtime on a single NVIDIA RTX 3080 GPU (with and without CUDA graph capture) on the left, and on CPU (AMD Ryzen 7 3700X) on the right. CUDA graph capture mostly eliminates the kernel launch overhead and thus isolates the raw computation time of the model. All graphs show the near-linear scaling in the number of tracked landmarks.

with position estimate $\mathbf{p}$, label $\mathbf{y}$, and $d = \sqrt{hw}$ of the labels' $h \times w$ bounding box. Assuming we take the RMS over landmarks for NAD (or equivalently, mean over landmarks for NAD^2), we drop the landmark index n . Then,

$$\text{NAD}^2(m) = \frac{1}{2(L-2m+1)} \sum_{l=1}^{L-2m+1} \|\bar{\mathbf{e}}_{l+m}(m) - \bar{\mathbf{e}}_l(m)\|_2^2 \quad (2)$$

(cf. [4], Eq. 6) with block averages $\bar{\mathbf{e}}(m)$ over m frames,

$$\bar{\mathbf{e}}_l(m) = \frac{1}{m} \sum_{i=l}^{l+m-1} \tilde{\mathbf{e}}_i. \quad (3)$$

A delta predictor (like QLOT) outputs the position increment $\Delta \mathbf{p}_l = \mathbf{p}_l - \mathbf{p}_{l-1}$ instead of the absolute position $\mathbf{p}$. Thus, we define the normalized error increment

$$\tilde{\mathbf{u}}_l = \tilde{\mathbf{e}}_l - \tilde{\mathbf{e}}_{l-1} = \frac{1}{d_l} (\Delta \mathbf{p}_l - \Delta \mathbf{y}_l) \quad (4)$$

when $d_l = d_{l-1}$ (a good approximation for successive frames). Note that NMF actually measures $\tilde{\mathbf{u}}_l$,

$$\text{NMF}_l = \frac{\|\mathbf{e}_l - \mathbf{e}_{l-1}\|_2}{d_l} = \|\tilde{\mathbf{e}}_l - \tilde{\mathbf{e}}_{l-1}\|_2 = \|\tilde{\mathbf{u}}_l\|_2 \quad (5)$$

and is aggregated in the RMS over frames. We can also show that $\text{NAD}(1) = \text{NMF}/\sqrt{2}$ since $\bar{\mathbf{e}}_l(1) = \tilde{\mathbf{e}}_l$ and

$$\text{NAD}^2(1) = \frac{1}{2(L-1)} \sum_{l=1}^{L-1} \|\tilde{\mathbf{e}}_{l+1} - \tilde{\mathbf{e}}_l\|_2^2 = \frac{\text{NMF}^2}{2}. \quad (6)$$

3.2. Ideal NAD Signatures

White jitter. Assume the position estimate decomposes into the label's trajectory plus white zero-mean estimation noise, $\mathbf{p} = \mathbf{y} + \boldsymbol{\varepsilon}_j$, whose pixel variance scales with face size, $\mathbb{E}\|\boldsymbol{\varepsilon}\|^2 = \sigma_j^2 d^2$. Then, with independent noise-free labels and scales, the normalized error $\tilde{\mathbf{e}}$ is white and has the same variance $\mathbb{E}\|\tilde{\mathbf{e}}\|^2 = \sigma_j^2$. As a result, successive block averages share no samples, and we obtain jitter of total variance (cf. [4], Eq. 17)

$$\text{NAD}^2(m) = \text{Var}(\bar{\mathbf{e}}(m)) = \sigma_j^2/m. \quad (7)$$

Independent label noise would add its own σ_{GT}^2/m term. A log-log NAD curve of pure white jitter therefore has a slope of $-1/2$.

Bias instability. Assume instead that the normalized error contains a slowly fluctuating bias with a power spectral density (PSD) of $S(f) \propto 1/f$ (so-called *flicker noise*) for $f \leq$

Process	Model / PSD	$\text{NAD}^2(m)$	Asymptotic log-log Slope
white jitter	$\mathbf{p} = \mathbf{y} + \boldsymbol{\varepsilon}_j$	σ_j^2/m	$-1/2$
bias instability	$S(f) \propto 1/f$	plateau $\approx (0.664B)^2$	0
free random walk	$\Delta \mathbf{p} = \Delta \mathbf{y} + \boldsymbol{\varepsilon}_u$	$\sigma_u^2(2m^2 + 1)/(6m)$	$+1/2$
linear drift	$\tilde{\mathbf{e}}_l = l \cdot \mathbf{v}$	$m^2 \ \mathbf{v}\ ^2/2$	$+1$
bounded walk, AR(1)	$\tilde{\mathbf{e}}_{l+1} = \rho \tilde{\mathbf{e}}_l + \boldsymbol{\varepsilon}_{l,d}$	Eq. (24)	$+1/2$ for $m \ll \tau_c$, $-1/2$ for $m \gg \tau_c$

Table 3. NAD signatures of canonical error processes. $\boldsymbol{\varepsilon}$ is zero-mean white noise of variance σ^2 . $\mathbf{v}$ is a non-zero constant.

f_0 and zero above (cf. [4], Eq. 18), with B^2 the flicker power density. In the time domain, such noise has no characteristic timescale. Its autocorrelation decays only logarithmically, and it is equivalently viewed as a superposition of AR(1)-type modes with time constants spread uniformly in log-time. Block-averaging m frames therefore reduces the variance only by $\log m$, and, as common-mode wander, is cancelled by NAD's block differences. The result is independent of m and yields the plateau of [4], Eq. 19,

$$\text{NAD}^2(m) = (2 \ln 2 / \pi) B^2 \approx (0.664B)^2. \quad (8)$$

Thus, a flat region (slope 0) in the log-log NAD plot indicates long-range correlated error that temporal averaging cannot reduce: the *bias instability floor*.

Free random walk. When the position increment decomposes into the label increment plus white zero-mean estimation noise, $\Delta \mathbf{p}_l = \Delta \mathbf{y}_l + \epsilon_l$, whose pixel variance scales with face size, $\mathbb{E} \|\epsilon_l\|^2 = \sigma_u^2 d_l^2$. Then, with noise-free label increments and independent scales, $\tilde{\mathbf{u}}_l = \tilde{\epsilon}_l / d_l$ is white with the same variance $\mathbb{E} \|\tilde{\mathbf{u}}_l\|^2 = \sigma_u^2$, and it is integrated into a free error random walk, $\tilde{\mathbf{e}}_l = \sum_{k \leq l} \tilde{\mathbf{u}}_k$. The difference of successive block averages is

$$\begin{aligned} \bar{\mathbf{e}}_{l+m}(m) - \bar{\mathbf{e}}_l(m) &= \frac{1}{m} \sum_{i=l}^{l+m-1} \sum_{k=i+1}^{i+m} \tilde{\mathbf{u}}_k \\ &= \frac{1}{m} \sum_{j=1}^{2m-1} w_j \tilde{\mathbf{u}}_{l+j} \end{aligned} \quad (9)$$

with triangular weights $w_j = \min(j, 2m - j)$. Since the increments are white, inserting Eq. (9) into Eq. (2) yields

$$\begin{aligned} \text{NAD}^2(m) &= \frac{\sigma_u^2}{2m^2} \sum_j w_j^2 = \frac{\sigma_u^2}{2m^2} \cdot \frac{m(2m^2 + 1)}{3} \\ &= \sigma_u^2 \frac{2m^2 + 1}{6m}, \end{aligned} \quad (10)$$

recovering the asymptote $\sigma_u^2 m / 3$ (cf. [4], Eq. 21, with $K^2 = \sigma_u^2$) and $\text{NAD}^2(1) = \sigma_u^2 / 2$. A log-log NAD curve of a free random walk therefore has an asymptotic slope of $+1/2$.

Linear drift. Assume a constant increment bias

$$\tilde{\mathbf{u}}_l = \mathbf{v}, \quad (11)$$

which integrates into a linear drift of the normalized error,

$$\tilde{\mathbf{e}}_l = \tilde{\mathbf{e}}_0 + l \cdot \mathbf{v}. \quad (12)$$

For a delta predictor, this is the failure mode of an uncompensated velocity bias. The process is deterministic and $\text{NAD}(m)$ measures it exactly for any clip length. Block averaging preserves the drift exactly,

$$\bar{\mathbf{e}}_l(m) = \tilde{\mathbf{e}}_0 + \left(l + \frac{m-1}{2}\right) \mathbf{v}, \quad (13)$$

so the difference of successive block averages is independent of the window position l ,

$$\bar{\mathbf{e}}_{l+m}(m) - \bar{\mathbf{e}}_l(m) = m\mathbf{v}, \quad (14)$$

giving (cf. [4], Eq. 23)

$$\text{NAD}^2(m) = \frac{m^2 \|\mathbf{v}\|^2}{2}. \quad (15)$$

$\text{NAD}^2(1) = \|\mathbf{v}\|^2 / 2$ and $\text{NMF}^2 = \|\mathbf{v}\|^2$. Temporal averaging is powerless against drift. The block difference grows linearly in m , the log-log slope $+1$ is the steepest signature in Table 3, and even a small bias $\mathbf{v}$ eventually dominates the considered error processes at large m .

Bounded walk, AR(1). Assume the normalized error reverts to zero as a stationary AR(1) process,

$$\tilde{\mathbf{e}}_{l+1} = \rho \tilde{\mathbf{e}}_l + \epsilon_{l,d}, \quad (16)$$

driven by zero-mean white innovations $\epsilon_{l,d}$ with $\mathbb{E} \|\epsilon_{l,d}\|^2 = \sigma_d^2$ (pixel variance $\sigma_d^2 d^2$, labels noise-free and scales independent, as before). Squaring the recursion and taking expectations gives the stationary variance

$$\sigma^2 = \mathbb{E} \|\tilde{\mathbf{e}}\|^2 = \sigma_d^2 / (1 - \rho^2), \quad (17)$$

and the autocovariance

$$\mathbb{E} \langle \tilde{\mathbf{e}}_l, \tilde{\mathbf{e}}_{l+h} \rangle = \sigma^2 \rho^{|h|} \quad (18)$$

decays with correlation time $\tau_c = -1 / \ln \rho$ frames ($\rho = e^{-1/\tau_c}$). Unlike the free random walk, the geometric pull toward zero dissipates integrated jitter as fast as the innovations inject it, so the error stays bounded at σ^2 regardless of the clip runtime L . In the delta domain, this restoring force appears as anticorrelated increments. Defining $\gamma_u(h)$ as the delta-autocovariance

$$\gamma_u(h) = \mathbb{E} \langle \tilde{\mathbf{u}}_l, \tilde{\mathbf{u}}_{l+h} \rangle, \quad (19)$$

at lag 1 this evaluates to

$$\gamma_u(1) = \mathbb{E} \langle \tilde{\mathbf{e}}_l - \tilde{\mathbf{e}}_{l-1}, \tilde{\mathbf{e}}_{l+1} - \tilde{\mathbf{e}}_l \rangle = -\sigma_d^2 \frac{1 - \rho}{1 + \rho}, \quad (20)$$

a fraction $-(1 - \rho)/2$ of the increment variance $\sigma_u^2 = 2\sigma_d^2/(1 + \rho)$. Each frame partially cancels the previous frame's delta error.

Furthermore, defining NAD in expectation,

$$\text{NAD}^2(m) = \frac{1}{2} \mathbb{E} \|\bar{\mathbf{e}}_{l+m}(m) - \bar{\mathbf{e}}_l(m)\|^2, \quad (21)$$

it can be shown to equal $[V(m) - C(m)]/m^2$ with

$$V(m) = \mathbb{E} \|m \bar{\mathbf{e}}_l\|^2 = \sigma^2 \left[m + 2 \sum_{h=1}^{m-1} (m-h) \rho^h \right], \quad (22)$$

the variance of a length- m block sum, and

$$C(m) = \mathbb{E} \langle m \bar{\mathbf{e}}_l, m \bar{\mathbf{e}}_{l+m} \rangle = \sigma^2 \rho \frac{(1 - \rho^m)^2}{(1 - \rho)^2}, \quad (23)$$

the covariance of two adjacent blocks. Putting it together,

$$\text{NAD}^2(m) = \frac{\sigma_d^2}{m^2(1-\rho^2)} \left[m + 2 \sum_{h=1}^{m-1} (m-h)\rho^h - \rho \frac{(1-\rho^m)^2}{(1-\rho)^2} \right]. \quad (24)$$

At $m = 1$ this gives exactly $\text{NAD}^2(1) = \sigma_d^2/(1+\rho)$. For slow zero-reversion ($\rho \rightarrow 1$), this expression measures the driving variance σ_d^2 directly. For $m \ll \tau_c$ (so $\rho^m \approx 1 - m/\tau_c$) the restoring force has not yet acted, the innovations integrate freely, and Eq. (24) reduces to the free-walk curve driven by $\epsilon_{l,d}$ with

$$\text{NAD}^2(m) \approx \sigma_d^2(2m^2 + 1)/(6m) \approx \sigma_d^2 m/3, \quad (25)$$

a $+1/2$ rise. For $m \gg \tau_c$ the block averages decorrelate ($\rho^m \rightarrow 0$) and

$$\text{NAD}^2(m) \approx \sigma^2 \frac{1+\rho}{(1-\rho)m} = \frac{\sigma_d^2}{(1-\rho)^2 m}, \quad (26)$$

a $-1/2$ tail whose level grows like τ_c^2 (since $1-\rho \approx 1/\tau_c$). Slower zero-reversion gives a later, higher hump but the same decay slope. The maximum sits at $m_c \approx 1.9\tau_c$ (numerical). The rising slope fitted over $[1, m_c]$ is damped relative to the free-walk benchmark, since the fit's lower endpoint is fixed at $m = 1$ while the branch rolls over at $m_c \propto \tau_c$.

3.3. Finite-Range Slope Benchmarks

The slopes of Table 3 are asymptotic. Least-squares fits over a finite range of m yield smaller effective slopes. For example, the exact random-walk curve

$$\text{NAD}(m) = \text{NAD}(1)\sqrt{(2m^2 + 1)/(3m)} \quad (27)$$

fitted over the range $m \in [1, 10]$ has slope $+0.43$, not $+1/2$. All fitted slopes in Table 4 are compared against finite-

range benchmarks computed from the exact curves over the reported branch ranges.

3.4. Estimation and Fitting Protocol

The per-clip relative error of $\text{NAD}(m)$ is

$$\sigma(\delta) = 1/\sqrt{2(L/m - 1)} \quad (28)$$

(cf. [4], Eq. 26). With $L = 120$ frames per clip: 21% at $m = 10$, 34% at $m = 22$, and 71% at $m = 60$. Averaging over 150 clips reduces this to 1.7%, 2.7%, 5.8%, and over 3 seeds (assuming approximately independent, homogeneous clips and seeds) to 1%, 1.6%, 3.3%, respectively. We therefore fit the clip and seed averaged curves.

Slopes are least-squares fits of $\log \text{NAD}(m)$ vs. $\log m$ over locally stable intervals. Two caveats: (i) ground-truth annotation noise adds an independent σ_{GT}^2/m term, damping apparent rises. The near-exact match of A1's rise to the free-walk benchmark shows this contribution is negligible. (ii) If m_c varies across clips, the averaged falling flank is a mixture of humps and its slope is biased toward 0. Per-clip curves confirm a large spread of peak locations (QLOT-final: median $m_c = 22$, IQR [11, 49]; A1: median 12, IQR [8, 26]), so the averaged tails in Table 4 are conservative (biased toward 0). The per-clip rises are themselves concave, i.e., each clip is closer to a single-timescale bounded walk.

3.5. Fitted Results and Interpretation

A1 integrates freely, then corrects late. Its rising slope matches the finite-range $m \in [1, 10]$ free random walk benchmark. Delta errors are essentially uncorrelated, and visual feedback engages only after ≈ 10 frames of free integration, consistent with a memoryless architecture whose corrections require the accumulated error to be visible in the current frame. A continuous (AR(1)-type) restoring force

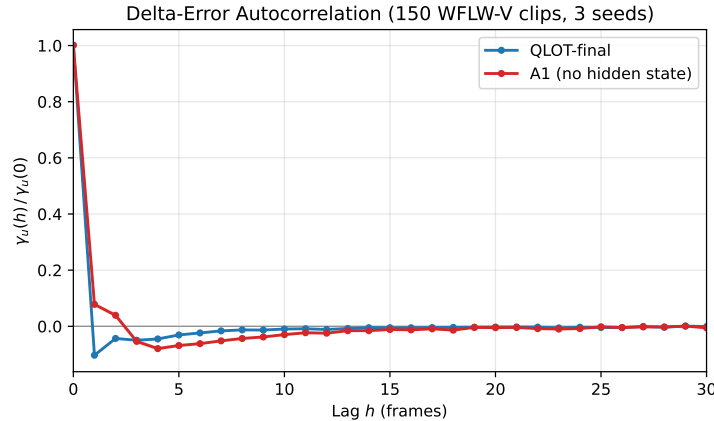


Figure 2. Autocorrelation of the face-size-normalized delta error $\tilde{\mathbf{u}}_l = \tilde{\mathbf{e}}_l - \tilde{\mathbf{e}}_{l-1}$ on WFLW-V (150 clips, mean over 3 seeds, seed-to-seed spread below the marker size). QLOT-final's lag-1 correlation is negative (-0.10). Each frame's update partially cancels the previous frame's delta error, which damps the NAD rise. A1's lag-1 correlation is slightly positive ($+0.08$) and turns negative only from lag 3 onward, so its correction is weaker and acts at longer lags.

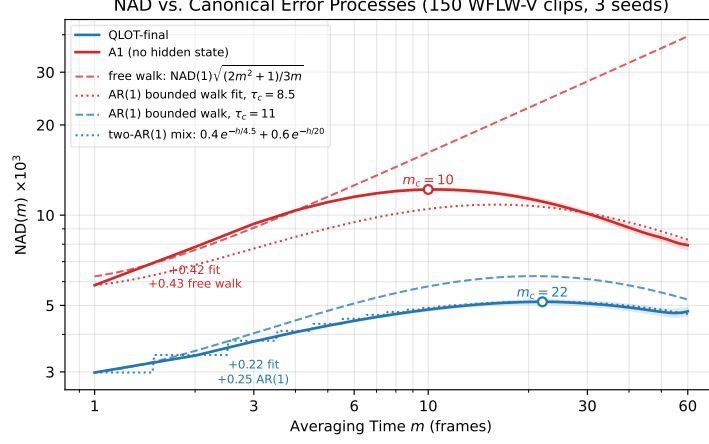


Figure 3. Measured NAD curves of A1 and QLOT-final (solid, shaded bands are the relative error of Eq. (28), divided by $\sqrt{150 \cdot 3}$) against the canonical error processes of Table 3 (dashed/dotted). Theory curves are anchored at the measured NAD(1), except A1’s free walk, which is scaled by 1.07 for a better rise fit. Correlation times and mixture weights are fit by grid search, minimizing log-space RMSE over $m \in [1, 60]$. A1’s best-fit AR(1) ($\tau_c = 8.5$) tracks the turnover but damps the rise to +0.27 (measured +0.42) and peaks too late ($m_c = 16$ vs. 10). A continuous restoring force is not supported for A1. QLOT-final’s AR(1) ($\tau_c = 11$) matches the damped rise (+0.25 benchmark over $[1, 22]$ vs. +0.22 measured) and the peak time ($m_c \approx 1.9\tau_c \approx 21$ vs. 22) but peaks too high and decays too steeply past the peak (−0.22 vs. −0.13). The two-AR(1) mixture $\gamma(h) = \sigma^2 [0.4e^{-h/4.5} + 0.6e^{-h/20}]$ reproduces the rise (+0.23), peak ($m_c = 22$), and flat tail (−0.12), consistent with the cross-clip spread of correlation times (per-clip m_c IQR $[11, 49]$).

would damp the rise to $\approx +0.25$ and is not supported by the data. The best-fit bounded walk ($\tau_c = 8.5$, Figure 3) damps the rise to +0.27 and peaks too late ($m_c = 16$ vs. measured 10), even though it tracks the turnover better than the free walk.

QLOT-final corrects immediately. Its rise, half the free-walk benchmark, is quantitatively matched by an AR(1) error with $\tau_c \approx 11$ frames (Figure 3). Temporal memory enables anticorrelated, self-correcting delta errors from the first frame onward. The later maximum ($m_c = 22$) marks a longer error correlation time: smaller and slower errors.

Delta-error autocorrelation supports this interpretation. The standardized, channel-averaged correlogram in Figure 2 shows a negative lag-1 correlation for QLOT-final (−0.10), whereas A1 is positive at lag 1 (+0.08) and first becomes negative at lag 3. Thus, QLOT-final’s normalized error increments tend to oppose the preceding increment immediately, while A1 exhibits a delayed correction signature. Because the figure normalizes each clip and channel before averaging, it is a qualitative check of correction timing

rather than a covariance-weighted decomposition of the NAD curve.

Both tails are flatter than mean reversion. The stable falling slopes (−0.36, −0.13) lie between the $-1/2$ of a single-timescale AR(1) and a flat curve, consistent with heterogeneous correlation times but not uniquely identifying bias instability. A two-timescale correlation $\gamma(h) = \sigma^2 [0.4e^{-h/4.5} + 0.6e^{-h/20}]$, one fast mode near the per-clip IQR lower edge and one slow mode near the median, reproduces QLOT-final’s rise, peak, and flat tail simultaneously (Figure 3). From $m_c = 22$ to $m = 60$, QLOT-final’s NAD decreases only from 5.14×10^{-3} to 4.78×10^{-3} , or 7%. Thus, temporal averaging provides limited additional attenuation over the measured range. The data do not establish an asymptotic floor or a hard bound for smoothing-based post-processing.

Negatives. Neither mean curve shows a +1 segment. No drift-dominated regime is detected within the measured range.

Model	Rise (fit)	Benchmark (branch)	Fall (fit)	m_c
A1	+0.42 ($m \in [1, 4]$)	+0.43, free walk ($m \in [1, 10]$)	−0.36 ($m \in [25, 60]$)	10
QLOT-final	+0.22 ($m \in [1, 9]$)	$\approx +0.25$, AR(1) ($m \in [1, 22]$); $\tau_c \approx 11$	−0.13 ($m \in [30, 55]$)	22

Table 4. Fitted NAD slopes and maxima. Empirical slopes are estimated over locally stable fit intervals, while benchmarks are evaluated over the full observed rising branch from $m = 1$ to m_c . Averaging clips with different peak locations smears the mean curve near m_c , so the fit intervals stop before the rising branch rolls over.

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